

Auteur: Ilyas Sefiani
Studierichting: Informatica
Oefeningenreeks 4A nr. 12.8

$$\begin{aligned} & \int \left(\frac{5^{x-1} + 4^{x+1}}{3^x} \right) dx \\ &= \int \left(\frac{\frac{5^x}{5} + 4 \cdot 4^x}{3^x} \right) dx \\ &= \frac{1}{5} \int \frac{5^x}{3^x} dx + 4 \int \frac{4^x}{3^x} dx \\ &= \frac{1}{5} \int \left(\frac{5}{3} \right)^x dx + 4 \int \left(\frac{4}{3} \right)^x dx \\ &= \frac{\left(\frac{5}{3} \right)^x}{5 \ln \left(\frac{5}{3} \right)} + \frac{4 \left(\frac{4}{3} \right)^x}{\ln \left(\frac{4}{3} \right)} + C \end{aligned}$$

References